

VL804 Analog Power IC Design

Lab Assignment-1

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3.3V Transistor Parameters

Model File:

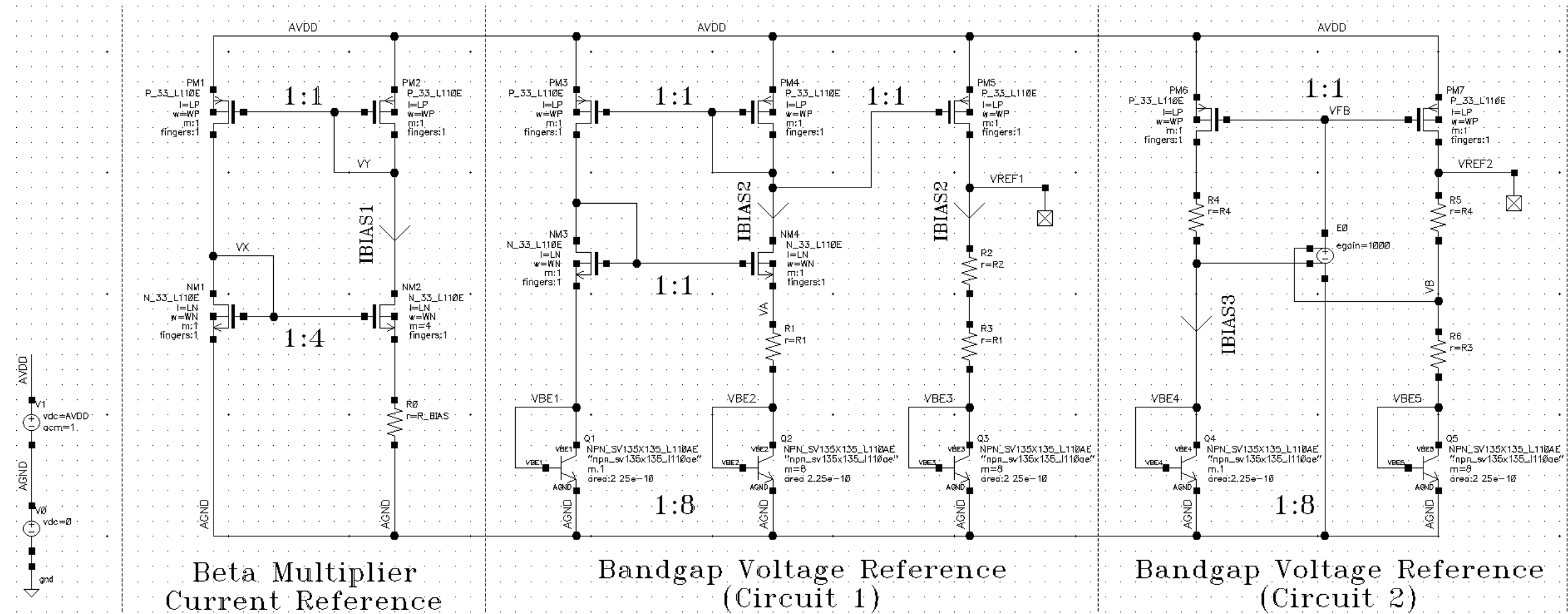
/Cadence_Work_110nm/_G-01-LOGIC_MIXED_MODE11-1P8M-

MMC_AL_L110AE/Designkits/Cadence_IC6/umc110ae/./Models/Spectre/l110ae_33_v102.lib.scs

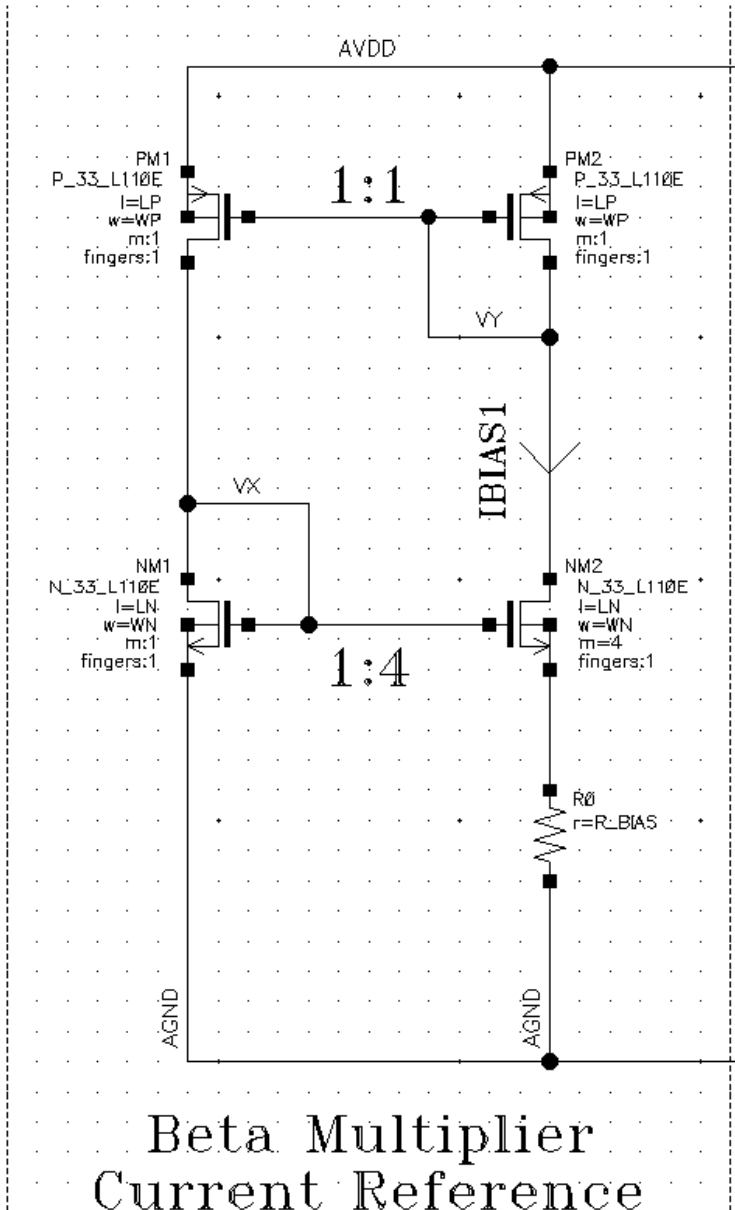
	Unit	N_33_L110E	P_33_L110E
L	um	8	8
Tox	nm	8	8
Cox	fF/um ²	4.60	4.60
u	M ² /V/s	0.0363	0.0097
Vth	V	0.6	0.6
Lambda	V ⁻¹	0.0110	0.0065
Kfit		0.7733	0.7733

Physical Constants		
Permittivity of free space	F/m	8.854E-12
Relative permittivity of SiO2		3.9
Relative permittivity of Si		11.7
Boltzmann constant, (k)	J/K	1.38E-23
Charge of electron, (q)	C	1.602E-19

Assignment Test-Bench (Schematic)

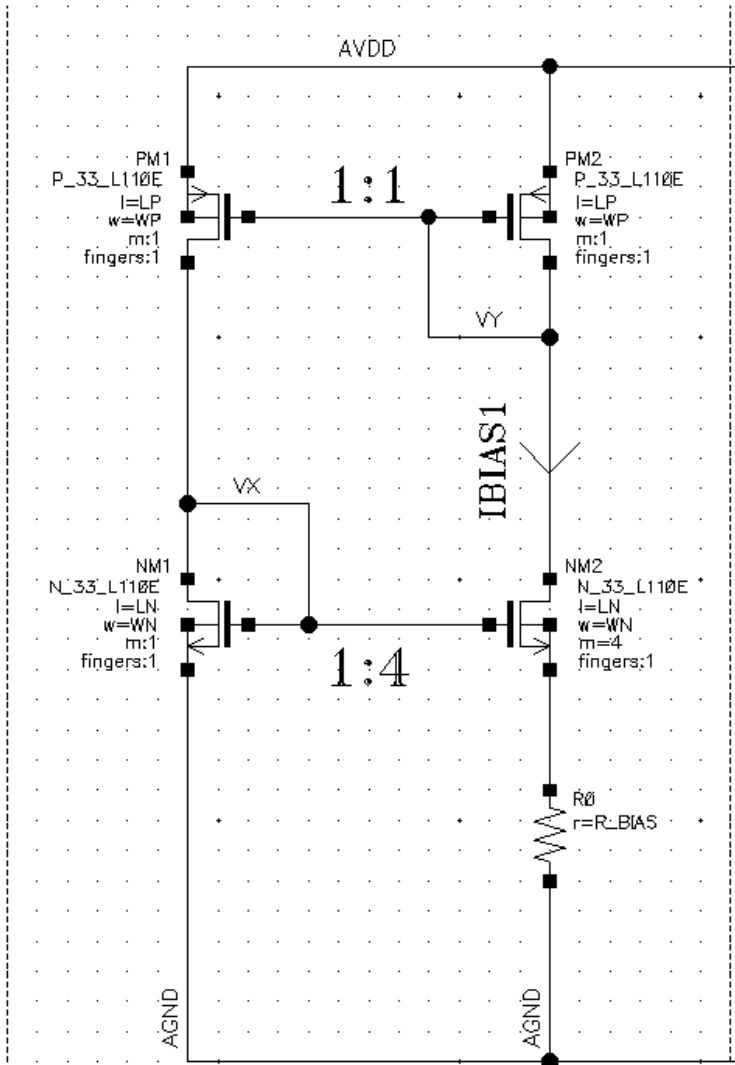


Beta Multiplier Current Reference



- Use “analogLib” resistors and capacitors.
- Given $AVDD = 3.0\text{ V}$, Over-drive voltage (V_{OV}) of 0.2 V for transistors, and $L = 8\text{ }\mu\text{m}$.
- Calculate the value of “R_BIAS”, “WN”, & “WP” for $IBIAS = 100\text{ nA}$.
Use $\beta = [K_{fit} * \mu * C_{ox} * (W/L)]$ to calculate “R_BIAS”.
- Perform DC operating point simulation with a temperature sweep from -40°C to 150°C and evaluate the following,
 - Plot IBIAS versus temperature across $AVDD = 2.4\text{V}$, 3.0V , 3.6V , and slow, typical, fast process corners
 - V_{TH_NM1} @ -40°C , 27°C , and 125°C
 - Gm_NM1 @ -40°C , 27°C , and 125°C
 - IBIAS @ 27°C
 - Average slope of IBIAS with respect to temperature

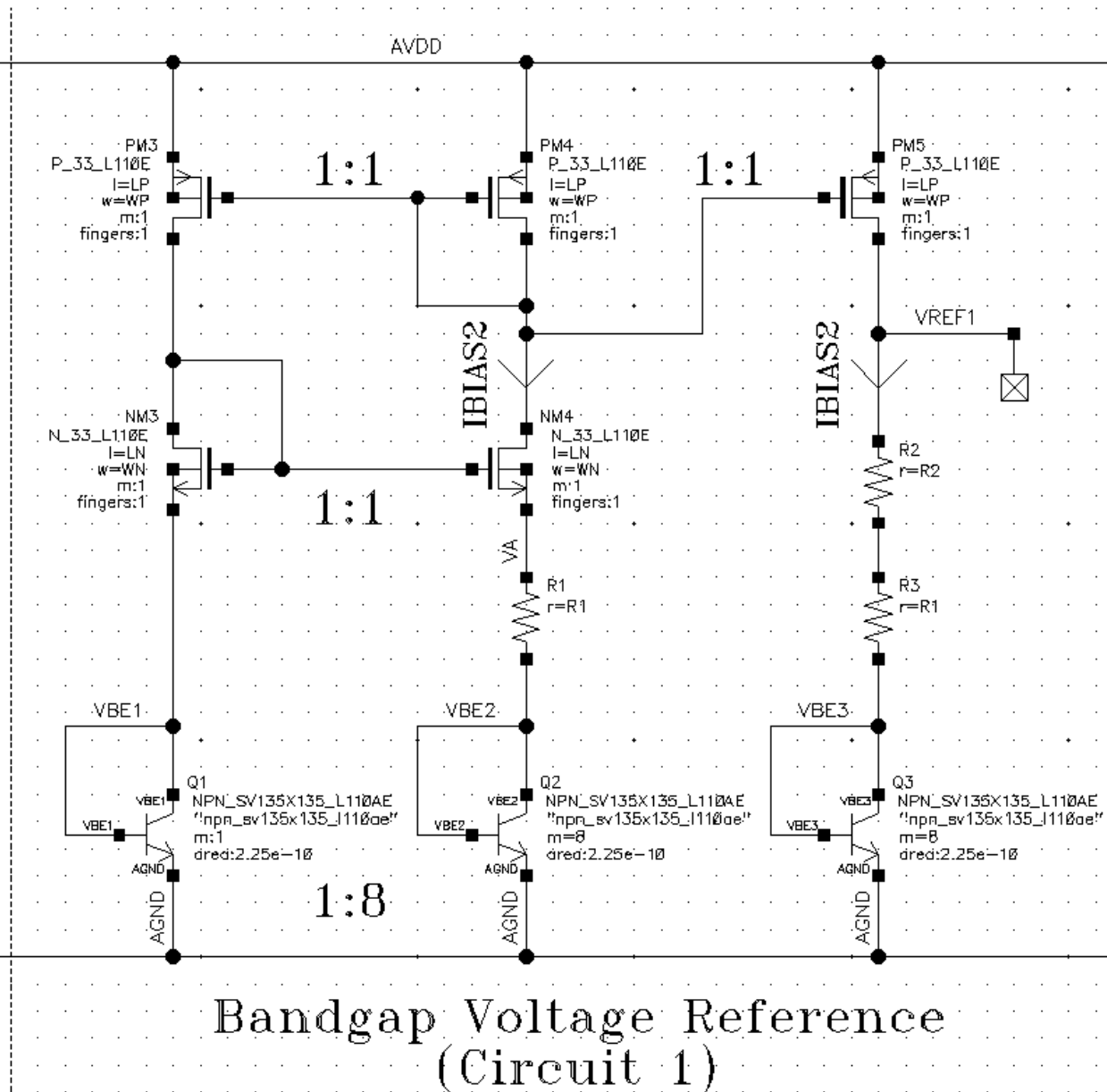
Beta Multiplier Current Reference

Beta Multiplier
Current Reference

- Comment on the results of Gm_NM1 and IBIAS across PVT corners

[illegible]

Bandgap Voltage Reference (Circuit 1)



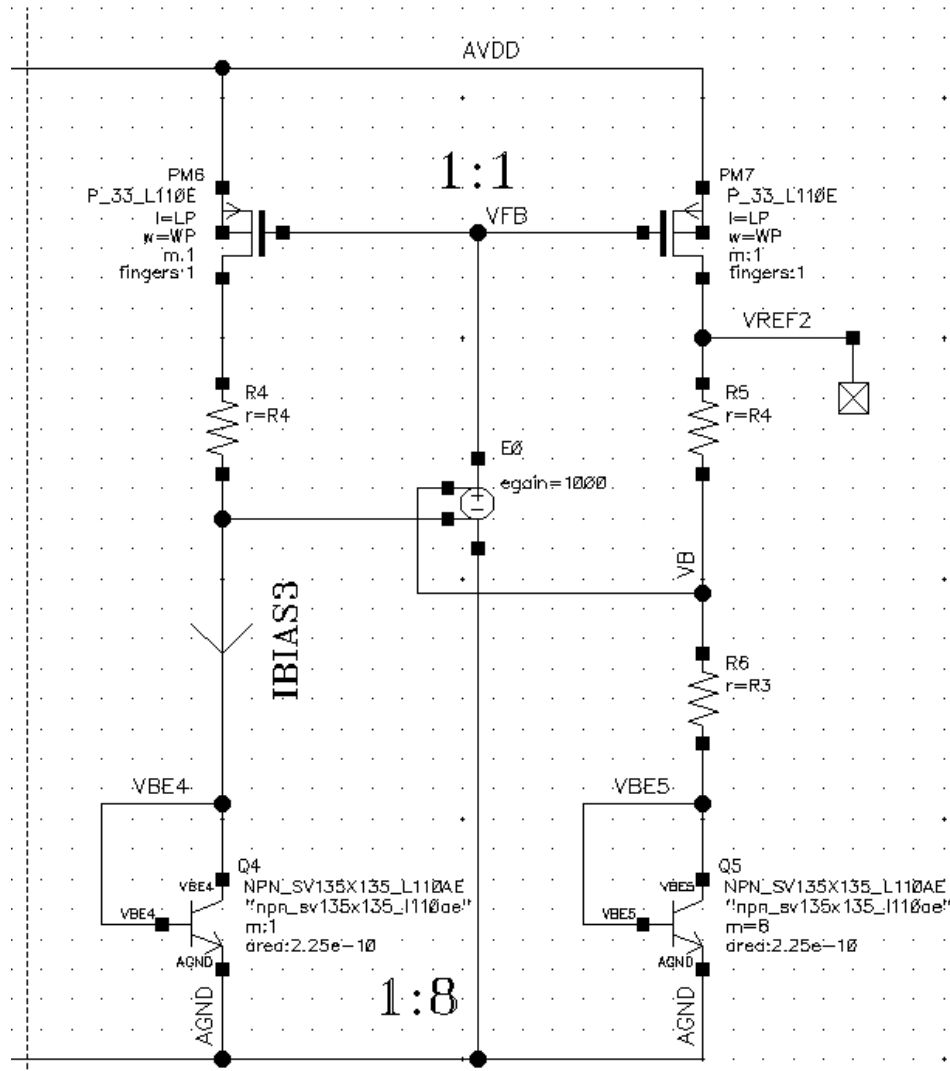
- Use “analogLib” resistors and capacitors.
- Given $AVDD = 3.0\text{ V}$, Over-drive voltage (V_{OV}) of 0.2 V for transistors, and $L = 8\text{ }\mu\text{m}$.
- Calculate the value of “R1” ($R1 = V_T \cdot \ln(8) / I_{BIAS1}$), “WN”, & “WP” for $I_{BIAS1} = 100\text{ nA}$.
 - Calculate the value of “R1”
 - $R2 = R1 \cdot [((dV_{BE3}/dT) \cdot Q/K / \ln(8)) - 1]$
 - Adjust the value of R2, to adjust for the errors in $(dV_{REF1}/dT) = 0$ due to channel length modulation effect and other non-idealities.

Bandgap Voltage Reference (Circuit 1)

- Perform DC operating point simulation with a temperature sweep from -40°C to 150°C and evaluate the following,
 - Plot VREF1 across PVT corners

[illegible]

Bandgap Voltage Reference (Circuit 2)



Bandgap Voltage Reference
(Circuit 2)

- Use “analogLib” resistors and capacitors.
- Given $AVDD = 3.0\text{ V}$, Over-drive voltage (V_{OV}) of 0.2 V for transistors, and $L = 8\text{ }\mu\text{m}$.
- Calculate the value of “R3” ($R3 = VT \cdot \ln(8) / IBIAS2$), “WN”, & “WP” for $IBIAS2 = 100\text{ nA}$.
 - Calculate the value of “R3”
 - $R4 = R3 \cdot [((dVBE5/dT) \cdot Q/K / \ln(8)) - 1]$
 - Adjust the value of R4, to adjust for the errors in $(dVREF2/dT) = 0$ due to channel length modulation effect and other non-idealities.

Bandgap Voltage Reference (Circuit 2)

- Perform DC operating point simulation with a temperature sweep from -40°C to 150°C and evaluate the following,
 - Plot VREF2 across PVT corners

[illegible]